GYNECOLOGY

Genetic contribution of reproductive traits to risk of uterine leiomyomata: a large-scale, genome-wide, cross-trait analysis

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BACKGROUND: Although phenotypic associations between female reproductive characteristics and uterine leiomyomata have long been observed in epidemiologic investigations, the shared genetic architecture underlying these complex phenotypes remains unclear.

OBJECTIVE: We aimed to investigate the shared genetic basis, pleiotropic effects, and potential causal relationships underlying reproductive traits (age at menarche, age at natural menopause, and age at first birth) and uterine leiomyomata.

STUDY DESIGN: With the use of large-scale, genome-wide association studies conducted among women of European ancestry for age at menarche (n=329,345), age at natural menopause (n=201,323), age at first birth (n=418,758), and uterine leiomyomata ($n_{cases}/n_{controls}$=35,474/267,505), we performed a comprehensive, genome-wide, cross-trait analysis to examine systematically the common genetic influences between reproductive traits and uterine leiomyomata.

RESULTS: Significant global genetic correlations were identified between uterine leiomyomata and age at menarche (r_g, −0.17; $P=3.65\times10^{-10}$), age at natural menopause (r_g, 0.23; $P=3.26\times10^{-07}$), and age at first birth (r_g, −0.16; $P=1.96\times10^{-06}$). Thirteen genomic regions were further revealed as contributing significant local correlations ($P<.05/2353$) to age at natural menopause and uterine leiomyomata. A cross-trait meta-analysis identified 23 shared loci, 3 of which were novel. A transcriptome-wide association study found 15 shared genes that target tissues of the digestive, exo- or endocrine, nervous, and cardiovascular systems. Mendelian randomization suggested causal relationships between a genetically predicted older age at menarche (odds ratio, 0.88; 95% confidence interval, 0.85−0.92; $P=1.50\times10^{-10}$) or older age at first birth (odds ratio, 0.95; 95% confidence interval, 0.90−0.99; P=.02) and a reduced risk for uterine leiomyomata and between a genetically predicted older age at natural menopause and an increased risk for uterine leiomyomata (odds ratio, 1.08; 95% confidence interval, 1.06−1.09; $P=2.30\times10^{-27}$). No causal association in the reverse direction was found.

CONCLUSION: Our work highlights that there are substantial shared genetic influences and putative causal links that underlie reproductive traits and uterine leiomyomata. The findings suggest that early identification of female reproductive risk factors may facilitate the initiation of strategies to modify potential uterine leiomyomata risk.

Key words: age at first birth, age at menarche, age at natural menopause, causal relationship, genetic correlation, pleiotropic effect, uterine leiomyomata

Introduction

Uterine leiomyomata (UL), or fibroids, are benign gynecologic tumors that affect nearly 70% of women of reproductive age.[1] Asymptomatic in most cases, UL nevertheless cause infertility, pelvic pain, and severe menorrhea in a substantial proportion of women, frequently leading to surgical treatments.[2] Although the pathogenesis of UL remains to be elucidated fully, the increasing incidence of diagnosed UL during reproductive years and the decreasing incidence with menopause implicate hormonal changes related to menstruation and pregnancy.[2]

Observational studies have consistently identified that female reproductive characteristics, such as age at menarche (AAM) and age at first birth (AFB), are linked to the risk of UL. For example, the longitudinal Nurses' Health Study II involving 95,061 women reported inverse associations between AAM and UL (≥16 years: relative risk [RR], 0.68; 95% confidence interval [CI], 0.53−0.88) and between AFB and UL (≥25 years: 10%−30% risk reduction when compared with ≤24 years),[3] both of which were later replicated in the same cohort using a 3-time augmented sample size of cases (AAM ≥16 years: RR, 0.77; 95% CI, 0.68−0.88; AFB ≥31 years: RR, 0.71; 95% CI, 0.64−0.79).[4] Another cohort study of 22,895 African-American women reported similar associations (AAM ≥15 years: RR, 0.70; 95% CI, 0.50−0.80; AFB ≥ 30 years: RR, 0.60; 95% CI, 0.40−0.90).[5] Nevertheless, conclusions regarding causality cannot be drawn easily from observational designs because of the possibility of bias and latent confounding, especially for a condition with such a high incidence and a great tendency to be asymptomatic.[1] In contrast to the well-documented results for AAM and AFB, observational evidence on the timing of menopause (age at natural menopause [ANM]) remains scarce despite its critical role in determining the lifelong exposure of a woman to steroid sex hormones.[6]

One way of understanding the relationships among entangled traits is to investigate their genetic link, which may help to bridge the limitations of

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AJOG at a Glance

Why was this study conducted?

Little is known regarding the shared genetic architecture that underlies reproductive characteristics and uterine leiomyomata (UL). This study evaluated the shared genetic basis, pleiotropic effects, and potential causal relationships between reproductive traits, including age at menarche, age at natural menopause, and age at first birth, and the presence of UL.

Key findings

Findings support substantial shared genetic influences and putative causal links that explain the shared heritability between reproductive traits and UL.

What does this add to what is known?

This study highlights an intrinsic link that underlies reproductive traits and UL and sheds new light on the biologic mechanisms. Findings of causal relationships support the early identification of female reproductive risk factors for UL prevention.

observational study and provide valuable information on the intrinsic mechanisms. The pathogenesis of both female reproductive traits and UL is complex and involves polygenic risk factors. Traditionally, genetic investigations of UL have primarily focused on somatic rearrangement with a particular emphasis on identifying key driver variations. However, the emergence of genome-wide association studies (GWAS) has led to a paradigm shift in research approaches in this field. Recent GWAS that examined reproductive traits and UL separately have estimated that single nucleotide polymorphism (SNP)-heritability varied from 9% to 38% for reproductive traits[7–9] and was 13% for UL,[10] indicating a considerable genetic contribution. When reproductive traits and UL were examined together, complex genetic relationships were revealed as evidenced by substantial genome-wide genetic correlations reported in a previous study (AAM-UL: r_g, −0.16; ANM-UL: r_g, 0.17; AFB-UL: r_g, −0.14).[10] However, the extent and nature of such genetic relationships remain unclear. As essential markers of the onset and cessation, respectively, of ovarian and related endocrine activity, menarche and menopause play nontrivial roles in UL etiology. Understanding the influence of AFB on UL growth may inform prevention strategies for UL by managing modifiable reproductive behaviors.

Therefore, in this study, we performed a large-scale, genome-wide, cross-trait analysis to systematically examine the shared genetic architecture that underlies the observed correlation between female reproductive traits (AAM, ANM, AFB) and UL. Specifically, with the use of large-scale genetic data and advanced statistical genetics tools, we quantified the shared genetic basis, pleiotropic effects, and causal relationships of UL with AAM, ANM, and AFB. The overall study design is shown in Figure 1.

Materials and Methods

This study was conducted with the use of summary statistics from large-scale GWAS that have been conducted to date for each trait. To reduce potential bias from population stratification, all data were restricted to European populations.

Genome-wide association studies summary data

Reproductive traits

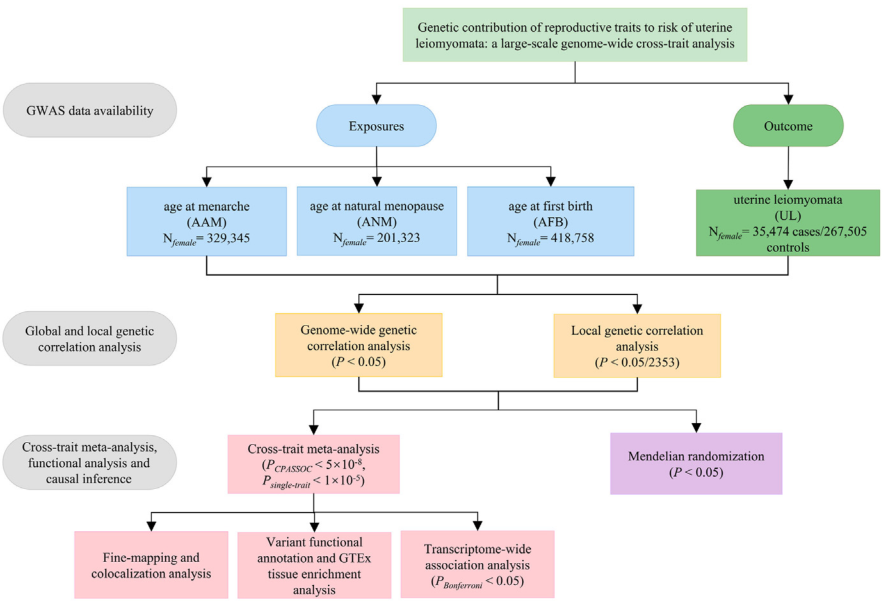
Three reproductive traits that were demonstrated to have polygenic components by previous GWAS were included. GWAS summary data for AAM were obtained from a meta-analysis of the Reproductive Genetics Consortium, 23andMe, and the United Kingdom Biobank and comprised 329,345 women.[7] GWAS summary data for ANM were obtained from a meta-analysis of the 1000 Genomes imputed studies, the Breast Cancer Association Consortium, and the United Kingdom Biobank and comprised 201,323 women.[8] Female-specific GWAS summary data for AFB were obtained from the Human Reproductive Behaviour Consortium meta-analysis that comprised 418,758 women.[9] For AAM and ANM, we selected 354 and 262 significant ($P<5\times10^{-08}$), independent, genome-wide autosomal SNPs, respectively, that were reported in the original GWAS to be instrumental variables (IVs). Because the original GWAS of AFB focused only on genetic associations unique to each sex (eg, female-specific lead SNP defined as having no effect in males yielding one single instrument), we applied clumping within a 500kb window (r^2 >0.001) on female-specific summary data and identified 93 independent, genome-wide, significant ($P<5\times10^{-08}$) SNPs. These index SNPs were used as IVs. We also retrieved a full set of summary statistics for the 3 reproductive traits for additional analysis.

Uterine leiomyomata

For UL, a large-scale GWAS was conducted by using meta-analyses data from 5 participating cohorts of the Fibro-GENE consortium (Women's Genome Health Study, Northern Finnish Birth Cohort, QIMR Berghofer Medical Research Institute, United Kingdom Biobank, and 23andMe) in 2019[10] and included 35,474 women with UL and 267,505 female controls. The UL cases and controls were identified based on either self-report (yes or no) or clinical documentation (hospital-linked medical records). The top-associated SNPs in the combined meta-analysis that reached a P threshold of 5×10^{-08} were reported. We extracted relevant information from the 27 GWAS-identified UL-associated significant autosomal SNPs and used those SNPs as IVs. We also retrieved a full set of summary statistics of UL (with the inclusion of data from 23andMe).

Details on the characteristics of each set of IVs are shown in Supplemental Tables 1 to 4. A table summarizing the

FIGURE 1
Overall study design of genome-wide, cross-trait analysis



We first retrieved summary statistics from large-scale GWAS conducted for age at menarche, age at natural menopause, age at first birth, and uterine leiomyomata, all in European ancestry. Second, we conducted global and local genetic correlation analyses to quantify the shared genetic basis. Third, we performed a cross-trait meta-analysis to identify pleiotropic variants, followed by transcriptome-wide association studies to detect shared expression-trait associations. Finally, we conducted a bidirectional Mendelian randomization analysis to evaluate potential causal relationships.

GTEX, Genotype-Tissue Expression project; *GWAS*, genome-wide association studies.

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information of all included GWASs is shown in Supplemental Table 5. For all analyses, the human reference genome build 37 (or hg19) was used.

Statistical analysis

We used a range of cutting-edge statistical approaches to investigate the genetic contribution of reproductive traits to the risk of UL. We first estimated the global genetic correlation analysis using linkage disequilibrium (LD) score regression (LDSC)[11] and calculated pairwise local genetic correlations using SUPERGNOVA.[12] These analyses enabled us to quantify the shared genetic basis between reproductive traits and UL, which were further decomposed into pleiotropy and causality. In assessing pleiotropy, we conducted a cross-trait meta-analysis using Cross-Phenotype Association (CPASSOC)[13] to identify shared loci affecting both reproductive traits and UL, followed by fine mapping,[14] colocalization,[15] and tissue enrichment analyses[16] to gain deeper biologic insights into the shared genetic components. Furthermore, we performed transcriptome-wide association studies (TWAS) using FUSION[17] to identify genes whose expression pattern across tissues implicate etiology or biologic mechanisms shared by reproductive traits and UL. Finally, we performed a bidirectional, 2-sample Mendelian randomization (MR) analysis to estimate the causal link between reproductive traits and UL. For more detailed information, please refer to the Methods section in the Supplemental Materials.

Results

Global and local genetic correlation analysis

A significant negative genomic correlation was found between AAM and UL (r_g, −0.17; P=3.65×10^{-10}) and between AFB and UL (r_g, −0.16; P=1.96×10^{-06}), all withstanding Bonferroni correction. We also identified a significant positive genomic correlation between ANM and UL (r_g, 0.23; P=3.26×10^{-07}) (Figure 2, A).

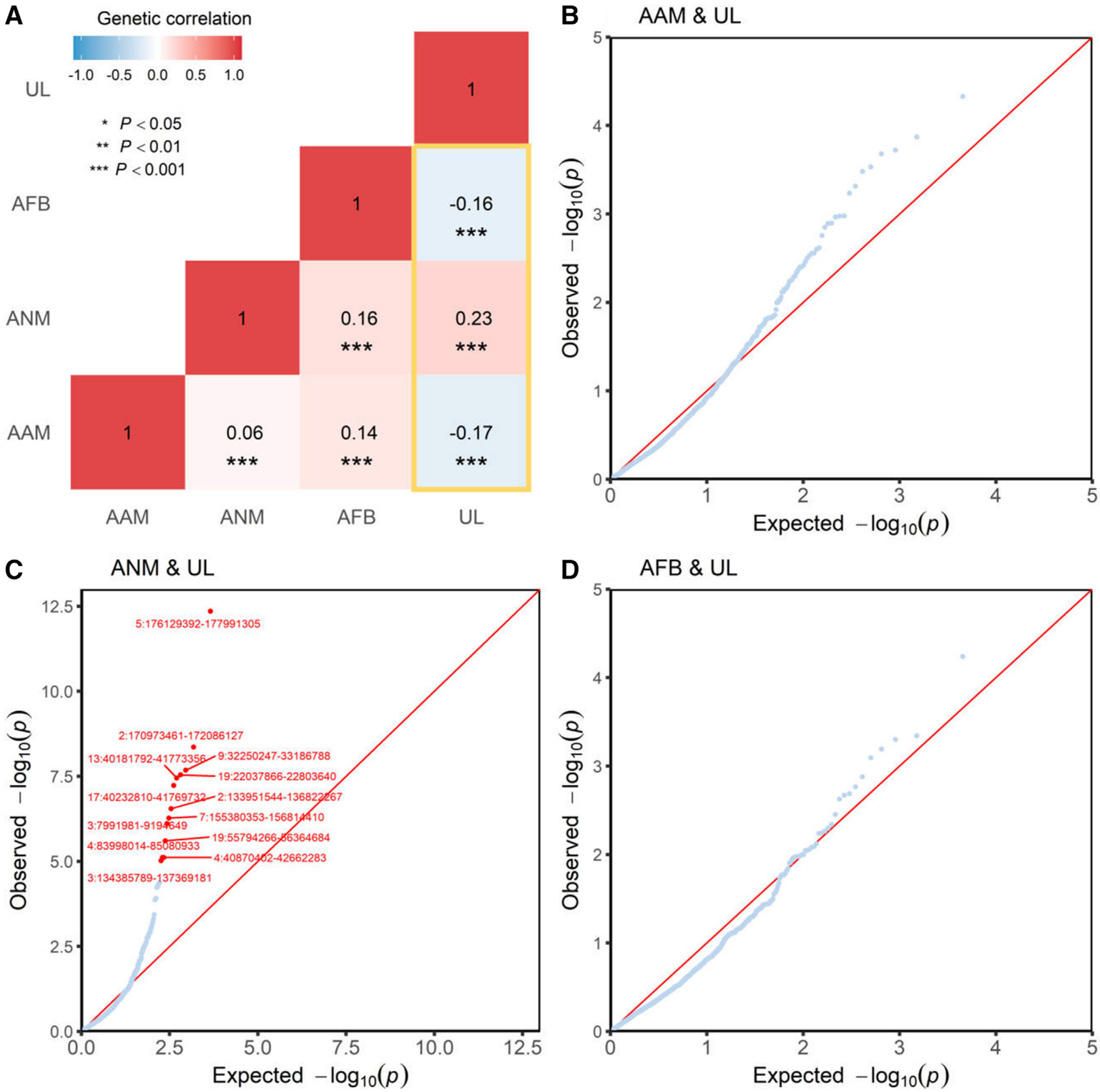
When exploring local genetic correlations, we observed significant local signals (P<.05/2353) for UL with ANM at 13 genomic regions (2q21.2-2q22.1, 2q31.1, 3p26.1-3p25.3, 3q22.2-2q22.3, 4p14-4p13, 4q21.22-4q21.23, 5q35.2-5q35.3, 7q36.3, 9p21.1, 13q14.11, 17q21.2-17q21.31, 19p12, and 19q13.42-19q13.43). Among these signals, 5q35.2-5q35.3 (chromosome 5: 176129392-177991305) represents the most significant region (P=4.45 × 10^{-13}), harboring the gene *UIMC1* that is known to be associated with both ANM[18] and UL.[10] The second most significant region (P=4.44 × 10^{-09}), 2q31.1 (chromosome 2: 170973461-172086127), harbors the gene *GAD1*, which is known to be implicated in the levels of sex hormone-binding globulin (SHBG).[19] The third most significant region (P=2.10 × 10^{-08}), 9p21.1 (chromosome 9: 32250247-33186788), harbors gene *APTX* that was previously reported to be associated with ANM.[18] No marked local signal was identified for AAM or AFB (Figures 2, B—C).

Cross-trait meta-analysis and pleiotropic loci

Motivated by the substantial genetic overlap, we further performed pairwise CPASSOC to identify pleiotropic loci. In total, 23 independent loci reached genome-wide significance in CPASSOC ($P_{CPASSOC}$ <5×10^{-08} and $P_{single\text{-}trait}$ <1×10^{-05}), including 6 loci that were shared between AAM and UL, 13 loci that were shared between ANM and UL, and 6 loci that were shared between AFB and UL (Table 1).

After excluding SNPs that were in LD (r^2 ≥0.20) with previously reported single-trait—associated significant SNPs, we identified 2 novel SNPs (rs1260326 and rs7921352) at 2p23.3 and 10q23.31 that were shared between AAM and UL. SNP rs1260326 ($P_{CPASSOC}$=4.13×10^{-11}) at 2p23.3 was located near *GCKR*, a gene encoding a glucokinase regulator previously

FIGURE 2
Global and local genetic correlations between reproductive traits and uterine leiomyomata



The heatmap (**A**) indicates global genetic correlations, the magnitude of which is represented by color. *Red*, positive genetic correlation; *blue*, negative genetic correlation. Darker colors correspond to stronger correlations. In the QQ plots (**B—D**), *red points* represent genomic regions that contribute significant local genetic correlations as estimated by SUPERGNOVA (P <0.05/2353).

AAM, age at menarche; *AFB*, age at first birth; *ANM*, age at natural menopause; *UL*, uterine leiomyomata.

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implicated in SHBG levels.[20] SNP rs7921352 ($P_{CPASSOC}$=1.39×10^{-10}) at 10q23.31 was located near *RNLS*, a gene encoding a flavin adenine dinucleotide-dependent amine oxidase associated with chronic kidney disease,[21] type 1 diabetes,[22] and various types of cancer.[23,24] For ANM and UL, 1 novel shared SNP (rs6430580) at 2q21.3 was found, located at *R3HDM1*, a previously reported potential oncogene.[25] We identified no novel shared SNP for AFB and UL.

TABLE 1

Pleiotropic SNPs between reproductive traits and uterine leiomyomata identified in the cross-trait meta-analysis ($P_{CPASSOC}$ <5×10^{-08}, $P_{single\text{-}trait}$ <1×10^{-05})

Index SNP	A1/A2	Chr: position	Cytoband	BETA		*P*		$P_{CPASSOC}$	Linear closest genes[a]	Interacting genes[b]
				Trait	UL	Trait	UL			
Age at menarche and uterine leiomyomata										
rs11031005	T/C	11:30226356	11p14.1	−0.04	0.10	4.64×10^{-09}	1.39×10^{-14}	7.67×10^{-20}	——	*FSHB*
rs56186913	T/C	11:207698	11p15.5	−0.03	0.05	2.03×10^{-08}	2.27×10^{-08}	6.47×10^{-14}	*BET1L, RIC8A, AC069287.1, RP11-304M2.5*	*LOC653486*
rs5742915	T/C	15:74336633	15q24.1	−0.02	−0.04	7.09×10^{-08}	2.02×10^{-06}	5.36×10^{-12}	*PML*	*LOXL1-AS1*
rs3813498	T/C	6:108944165	6q21	−0.03	0.05	6.71×10^{-07}	1.08×10^{-06}	5.13×10^{-11}	*FOXO3, SUMO2P8*	*LINC00222, FOXO3*
rs1260326[c]	T/C	2:27730940	2p23.3	0.02	0.05	7.46×10^{-06}	1.41×10^{-07}	4.13×10^{-11}	*GCKR*	*IFT172*
rs7921352[c]	T/C	10:90186123	10q23.31	−0.02	0.04	1.65×10^{-06}	1.09×10^{-06}	1.39×10^{-10}	*RNLS*	——
Age at natural menopause and uterine leiomyomata										
rs16991615	A/G	20:5948227	20p12.3	1.06	0.10	<1.00×10^{-324}	5.17×10^{-09}	<1.00×10^{-324}	*MCM8, Y_RNA, RP5-967N21.7*	*TRMT6*
rs11740768	A/G	5:176339499	5q35.2	0.32	0.06	4.66×10^{-154}	2.13×10^{-10}	1.57×10^{-171}	*UIMC1*	*HK3*
rs11031005	T/C	11:30226356	11p14.1	−0.20	0.10	2.08×10^{-31}	1.39×10^{-14}	1.83×10^{-41}	——	*FSHB*
rs7986407	A/G	13:41179798	13q14.11	−0.07	−0.06	1.53×10^{-08}	1.26×10^{-11}	4.14×10^{-19}	*FOXO1*	*FOXO1, LINC00598*
rs11031812	T/C	11:32505205	11p13	0.08	−0.05	2.44×10^{-09}	1.21×10^{-07}	2.66×10^{-15}	——	*WT1-AS*
rs62110991	T/C	19:22260842	19p12	0.11	0.06	5.72×10^{-11}	2.39×10^{-07}	1.52×10^{-15}	*ZNF257*	*ZNF208, ZNF257*
rs717454	T/C	2:100022772	2q11.2	−0.08	−0.04	1.26×10^{-10}	6.50×10^{-06}	2.42×10^{-13}	*REV1, EIF5B, RP11-527J8.1*	*TXNDC9*
rs1717777	A/T	11:30416270	11p14.1	−0.10	0.06	1.81×10^{-07}	2.64×10^{-06}	4.34×10^{-12}	*MPPED2*	*MPPED2*
rs12599260	A/G	16:50093238	16q12.1	0.07	0.04	1.40×10^{-07}	7.13×10^{-06}	3.65×10^{-11}	*RP11-429P3.4, RP11-429P3.3*	*PAPD5*
rs2277163	A/G	9:827224	9p24.3	−0.12	0.10	8.66×10^{-06}	3.91×10^{-07}	1.32×10^{-11}	——	*DMRT1, DMRT3*
rs5750931	T/C	22:40694719	22q13.1	−0.06	−0.05	4.23×10^{-06}	1.18×10^{-06}	4.58×10^{-11}	*TNRC6B*	*ADSL*
rs6430580[c]	A/G	2:136321951	2q21.3	0.10	0.07	2.01×10^{-06}	8.85×10^{-07}	1.85×10^{-11}	*R3HDM1*	*R3HDM1*
rs78330906	T/C	11:30176985	11p14.1	0.26	−0.19	4.23×10^{-06}	7.02×10^{-06}	2.34×10^{-10}	——	*FSHB, KCNA4*
Age at first birth and uterine leiomyomata										
rs2235529	T/C	1:22450487	1p36.12	0.07	0.14	3.48×10^{-07}	2.55×10^{-30}	2.58×10^{-30}	*WNT4*	*C1QC*
rs11031006	A/G	11:30226528	11p14.1	0.09	−0.10	3.77×10^{-10}	1.42×10^{-14}	8.88×10^{-21}	——	*FSHB*
rs71444571	A/G	12:123771015	12q24.31	0.07	−0.06	1.09×10^{-07}	5.47×10^{-08}	3.93×10^{-13}	*SBNO1, RNA5SP375, RP11-282O18.6*	*SBNO1*

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(continued)

TABLE 1
Pleiotropic SNPs between reproductive traits and uterine leiomyomata identified in the cross-trait meta-analysis ($P_{CPASSOC}$ <5×10^{-08}, $P_{single\text{-}trait}$ <1×10^{-05})
(continued)

Index SNP	A1/A2	Chr: position	Cytoband	BETA		*P*				
				Trait	UL	Trait	UL	$P_{CPASSOC}$	Linear closest genes[a]	Interacting genes[b]
rs1759645	T/C	6:34194866	6p21.31	0.07	−0.06	1.18×10^{-06}	3.90×10^{-07}	3.08×10^{-11}	—	*C6orf1*
rs4344303	T/C	1:22336120	1p36.12	−0.06	−0.05	7.22×10^{-06}	5.84×10^{-06}	8.66×10^{-10}	*CELA3A, RNU6-776P, RN7SL186P*	*LINC00339*
rs5742915	T/C	15:74336633	14q24.1	−0.05	−0.04	4.40×10^{-06}	2.02×10^{-06}	2.07×10^{-10}	*PML*	*LOXL1-AS1*

A novel SNP was identified only if all following criteria were satisfied: (1) the SNP reached genome-wide significance ($P_{CPASSOC}$ <5×10^{-08}) cross-trait; (2) the SNP did not reach genome-wide significance (5×10^{-08} <$P_{single\text{-}trait}$ <1×10^{-05}) in both single-trait GWAS; (3) the SNP was not in LD (r^2 <0.20) with any of those previously reported genome-wide significant SNPs of single traits. Position is according to human genome build 37 (hg19).

A1, effect allele; *A2*, other allele; *Chr*, chromosome; *SNP*, single nucleotide polymorphism; *UL*, uterine leiomyomata.

[a] Linear closest genes of index SNPs were mapped using VEP; [b] 3-Dimentional interacting genes of index SNPs were mapped using 3DSNP; [c] Indicates novel SNPs.

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Detailed annotations of each index SNP are shown in Supplemental Tables 6 and 7.

Identification of causal variants and colocalization

For each of the 23 pleiotropic SNPs, we further identified a 99% credible set of causal SNPs, providing targets for future downstream experimental analysis (Supplemental Table 8). We found 133 candidate causal SNPs across all loci shared by AAM and UL, 554 candidate causal SNPs across all loci shared by ANM and UL, and 252 candidate causal SNPs across all loci shared by AFB and UL. Of note, SNP rs5742915, shared by both AAM and UL and AFB and UL, was identified as having a posterior probability of 1.00 in the 99% credible sets.

We also performed a colocalization analysis to determine whether the pleiotropic SNPs driving the associations in 2 traits were the same or different. A considerable number of shared loci, including 50% (3/6) of the loci shared by AAM and UL, 46% (6/13) of the loci shared by ANM and UL, and 67% (4/6) of the loci shared by AFB and UL, colocalized at the same candidate SNPs (PPH_4 >0.7), reinforcing shared causal associations (Supplementary Table 9).

Biologic pathway and Genotype-Tissue Expression project tissue enrichment analysis

After multiple correction, gene ontology (GO) analysis across the shared AAM and UL genes revealed significant enrichments in the response of the interferon-alpha (GO: 0035455; P=4.40×10^{-06}) and interferon-beta (GO: 0035456; P=2.08×10^{-05}) pathways (Supplemental Table 10). Genotype-Tissue Expression ptoject (GTEx) tissue enrichment analyses identified that the expression of genes shared between AFB and UL was substantially enriched in the pancreas and left heart ventricle, all withstanding Bonferroni correction (Supplemental Figure 1). No additional pathway or tissue enrichment was found.

Transcriptome-wide association studies

Results from tissue-specific TWAS revealed gene-level genetic overlap

TABLE 2

TWAS significant genes shared between reproductive traits and uterine leiomyomata across 48 GTEx tissues (version 7) after conditional and joint analysis

Gene	Tissue_type	CHR	Trait		UL	
			BEST.GWAS.ID	$P_{Bonferroni}$	BEST.GWAS.ID	$P_{Bonferroni}$
Age at menarche and uterine leiomyomata						
ARL14EP	Testis	11	rs10835649	1.88×10^{-04}	rs11031006	9.54×10^{-09}
Age at natural menopause and uterine leiomyomata						
ABCB9	Esophagus_muscularis	12	rs1879380	4.41×10^{-29}	rs1060105	3.35×10^{-03}
AC074117.10	Artery_tibial	2	rs704795	1.52×10^{-21}	rs1260326	2.97×10^{-02}
ARL14EP	Adrenal_gland, artery_aorta, spleen, testis, thyroid	11	rs11031006	3.79×10^{-26} to 1.75×10^{-04}	rs11031006	9.54×10^{-09} to 2.57×10^{-02}
CDK2AP1	adipose_subcutaneous, adipose_visceral_omentum, brain_frontal_cortex_BA9, cells_transformed_fibroblasts, colon_transverse, heart_left_ventricle lung, pancreas, skin_not_sun_exposed_suprapubic, skin_sun_exposed_lower_leg, small_intestine_terminal_ileum, stomach	11	rs1879380	3.32×10^{-28} to 2.64×10^{-19}	rs1060105	4.48×10^{-03} to 3.27×10^{-02}
ELF1	Brain_cortex	13	rs7986407	4.90×10^{-02}	rs7986407	6.90×10^{-03}
FGFR4	Adipose_subcutaneous, brain_amygdala, brain_frontal_cortex_BA9	5	rs2292256	3.56×10^{-149} to 2.07×10^{-127}	rs353489	4.19×10^{-05} to 7.94×10^{-04}
RGS17P1	Cells_EBV-transformed_lymphocytes	13	rs7986407	6.57×10^{-04}	rs7986407	2.71×10^{-05}
RP11-282O18.3	Artery_aorta, artery_tibial, brain_anterior_cingulate_cortex_BA24	12	rs1879380	4.78×10^{-27} to 5.72×10^{-22}	rs1060105	4.15×10^{-03} to 4.10×10^{-02}
TPTE2P5	Cells_transformed_fibroblasts	13	rs7986407	8.72×10^{-04}	rs7986407	2.06×10^{-05}
SUGT1P3	Esophagus_muscularis	13	rs7986407	2.73×10^{-04}	rs7986407	5.16×10^{-03}
SETD8	Liver	12	rs1879380	7.26×10^{-25}	rs1060105	3.09×10^{-02}
UIMC1	Brain_cerebellar_hemisphere, colon_transverse, esophagus_mucosa, esophagus_muscularis, lung, nerve_tibial, pancreas, skin_not_sun_exposed_suprapubic, skin_sun_exposed_lower_leg, stomach, testis	5	rs2292256	2.59×10^{-148} to 5.80×10^{-85}	rs353489	4.19×10^{-05} to 2.69×10^{-02}
WT1	Brain_frontal_cortex_BA9	11	rs10734411	5.16×10^{-03}	rs11031731	1.39×10^{-07}
VAT1	Nerve_tibial	17	rs9911630	1.91×10^{-21}	rs9646416	3.56×10^{-02}
ZNF346	Esophagus_mucosa, thyroid	5	rs2292256	6.45×10^{-130} to 7.09×10^{-69}	rs353489	1.64×10^{-04} to 2.61×10^{-04}
Age at first birth and uterine leiomyomata						
ABCB9	Esophagus_muscularis	12	rs7964876	5.87×10^{-04}	rs1060105	3.35×10^{-03}
ARL14EP	Testis	11	rs11031006	6.03×10^{-03}	rs11031006	9.54×10^{-09}

Xiao. Genetic relationships between reproductive traits and uterine leiomyomata. Am J Obstet Gynecol 2024.

(continued)

TABLE 2
TWAS significant genes shared between reproductive traits and uterine leiomyomata across 48 GTEx tissues (version 7) after conditional and joint analysis *(continued)*

			Trait		UL	
Gene	Tissue_type	CHR	BEST.GWAS.ID	$P_{Bonferroni}$	BEST.GWAS.ID	$P_{Bonferroni}$
CDK2AP1	Adipose_subcutaneous, brain_frontal_cortex_BA9, cells_transformed_fibroblasts, pancreas, small_intestine_terminal_ileum	12	rs7964876	7.16×10^{-03} to 4.00×10^{-02}	rs1060105	4.48×10^{-03} to 3.27×10^{-02}
RP11-282018.3	Artery_aorta	12	rs7964876	2.90×10^{-03}	rs1060105	4.10×10^{-02}

CHR, chromosome; *EBV*, Epstein-Barr virus; *GTEx*, Genotype-Tissue Expression project; *TWAS*, transcriptome-wide association studies; *UL*, uterine leiomyomata.
Xiao. Genetic relationships between reproductive traits and uterine leiomyomata. Am J Obstet Gynecol 2024.

(Supplemental Figure 2). After multiple correction, we identified a total of 15 significant ($P_{\text{Bonferroni}}$<.05), independent transcriptome-wide shared genes (including 1 shared by AAM and UL, 15 shared by ANM and UL, and 4 shared by AFB and UL) that were enriched in tissues of the digestive, exo- and endocrine, nervous, and cardiovascular systems (Table 2). Four of the 15 genes were located at pleiotropic loci identified in the cross-trait meta-analysis, including *CDK2AP1*, *UIMC1*, SETD8, and *ZNF346*.

Among the 15 TWAS-significant shared genes, 6 were previously reported to be susceptible to reproductive traits and/or UL (GWAS catalog accessed on April 24, 2023), including *ARL14EP* and *UIMC1* associated with AAM, ANM, and UL; *ZNF346* associated with ANM and UL; *VAT1* associated only with ANM; and *FGFR4* and *WT1* associated only with the presence of UL. We identified 9 novel genes. Although the biologic functions of *AC074117.10*, *RGS17P1*, *RP11-282O18.3*, *TPTE2P5*, and *SUGT1P3* remain unclear, the novel gene *ABCB9* encodes an antigen processing-like transporter involved in the development and progression of various malignant tumors.[26,27] *CDK2AP1* encodes a cyclin-dependent kinase 2 (CDK2)−associated protein that plays a role in both cell-cycle and epigenetic regulation.[28] *ELF1* encodes a transcription factor that plays a unique role in gene regulation in the endometrium.[29] *SETD8* encodes a histone methyltransferase that functions in human carcinogenesis.[30]

Mendelian randomization analysis of reproductive traits and uterine leiomyomata

MR analysis found convincing evidence to support a causal relationships between the 3 reproductive traits and the risk for UL. The random-effect inverse-variance weighted (IVW) approach was used (in both univariate MR and multivariable [MV] MR analyses) because of the presence of between-SNP heterogeneity ($P_{\text{Cochran's Q}}$<.05) (Supplemental Table 11). Using IVW, genetically predicted later AAM was significantly associated with a reduced risk of UL (odds ratio [OR], 0.88; 95% CI, 0.85−0.92; $P=1.50\times10^{-10}$). The estimates remained consistent in both MR-Egger regression (OR, 0.81; 95% CI, 0.73−0.90; $P=8.00\times10^{-5}$) and the weighted median approach (OR, 0.89; 95% CI, 0.84−0.94; $P=1.20\times10^{-5}$). Similar associations were found between genetically predicted AFB and UL risk (IVW OR, 0.95; 95% CI, 0.90−0.99; P=.02; MR-Egger regression OR, 0.80; 95% CI, 0.66−0.97; P=.03; weighted median OR, 0.95; 95% CI, 0.91−0.99; P=.03). In contrast, genetically predicted later ANM was significantly associated with an increased risk for UL (IVW OR, 1.08; 95% CI, 1.06−1.09; $P=2.30\times10^{-27}$; MR-Egger regression OR, 1.10; 95% CI, 1.08−1.13; $P=4.60\times10^{-13}$; weighted median OR, 1.08; 95% CI, 1.06−1.10; $P=6.50\times10^{-23}$) (Figures 3, A−C). No evidence of horizontal pleiotropy was observed, except for ANM (AAM MR-Egger intercept, 0.0034; P=.08; ANM MR-Egger intercept, −0.0044; P=.02; AFB MR-Egger intercept, 0.0110; P=.11). Sensitivity analyses supported the robustness of our primary results (Supplemental Figure 3). MVMR analyses generated consistent associations after adjusting for body mass index or the other reproductive factors (either in isolation or in combination), suggesting independent causal effects (Supplemental Figure 4).

No evidence for a causal association in the reverse direction was found; genetic liability for UL did not seem to influence AAM (IVW OR, 1.01; 95% CI, 0.97−1.05; P=.59), ANM (IVW OR, 1.26; 95% CI, 0.65−2.41; P=.49), or AFB (IVW OR, 0.97; 95% CI, 0.86−1.09; P=.60) (Figure 3, D).

Comment

Principal findings

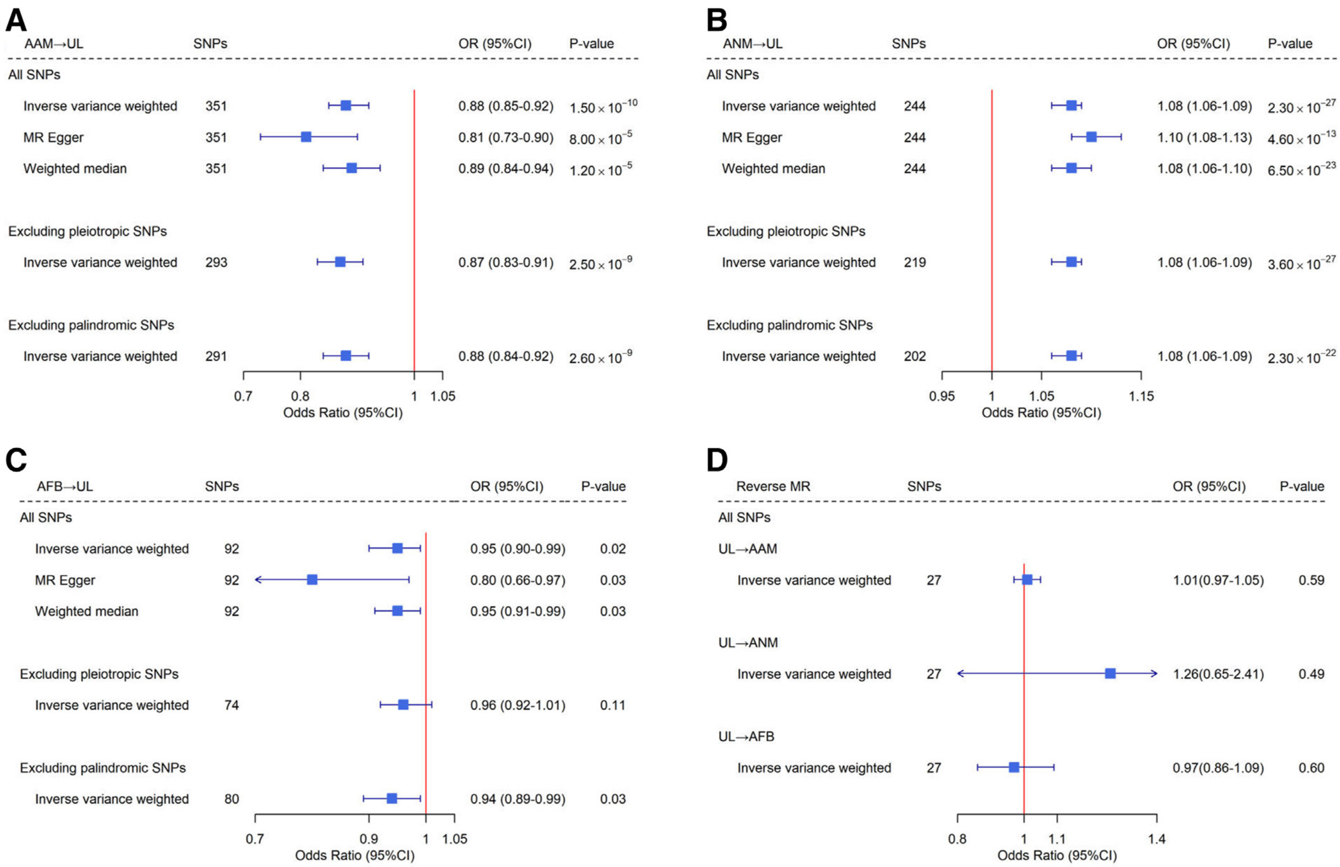
This was a comprehensive assessment of the genetic relationships between female reproductive characteristics and UL. We identified intrinsic links between UL and AAM, ANM, and AFB, evidenced by the considerable globally and locally shared heritability. By performing further CPASSOC and MR analyses, we found evidence that such genetic overlaps could be explained by both shared genetic loci and causal relationships.

Interpretation of study findings and comparison with existing literature

Overall, modest genetic correlations between AAM and UL, AFB and UL, and ANM and UL were identified at the genome-wide level. Although directionally consistent with genetic correlations

FIGURE 3

Bidirectional Mendelian randomization analysis between reproductive traits and uterine leiomyomata



Blue boxes denote point estimates of the causal effects and error bars denote 95% confidence intervals (CI).

AAM, age at menarche; *AFB*, age at first birth; *ANM*, age at natural menopause; *OR*, odds ratio; *MR*, Mendelian randomization; SNPs, single-nucleotide polymorphisms; *UL*, uterine leiomyomata.

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that were previously estimated with a substantially improved statistical power derived from the updated GWAS, we identified marked r_g between each pair of traits, including those that had reportedly failed to reach statistical significance in previous LDSC studies.[10] By partitioning the whole genome into LD-defined regions, we identified multiple common regional signals between ANM and UL with locally elevated heritability, which further reinforced a noteworthy shared genetic basis between reproductive characteristics and UL, specifically related to ANM. The lack of shared regional signals between AAM and UL and between AFB and UL could be attributed to the highly polygenic nature of the investigated traits or a consequence of the high test burden associated with local genetic correlation analysis.

A substantial shared genetic basis can reflect shared genetic influences in which specific genetic alleles confer risk to both traits or/and a directional (causal) relationship in which genetic alleles can influence one trait via their effects on the other.[31] In our downstream analysis performed to explore these options, 23 shared loci between female reproductive traits and UL were identified. Among these loci, 12 showed strong evidence of colocalization (PPH_4 >0.7), demonstrating etiologic connections. Notably, these shared loci harbor genes that were previously implicated in tumor growth (*RIC8A*, *PML*, *FOXO3*, *RNLS*, *MCM8*, *UIMC1*, *FOXO1*, *REV1*, *EIF5B*, *MPPED2*, *TNRC6B*, and *WNT4*),[32–42] cell proliferation or differentiation (*EIF5B*, *MPPED2*, *WNT4*, and *UIMC1*),[36,39,41,43] physiological metabolism (*PML* and *GCKR*),[20,33] and hormone secretion (*MCM8* and *UIMC1*).[35,44] Additional loci that did not attain genome-wide significance in single trait-specific GWAS can be observed using a cross-trait meta-analysis because of increased statistical power that can be achieved by combining multiple traits. Indeed, our cross-trait meta-analyses identified 3 novel loci (*GCKR*, *RNLS*, and *R3HDM1*) that were associated jointly with AAM and UL or ANM and UL.

GCKR, located at a novel shared locus (2p23.3) identified for AAM and UL, encodes a glucokinase regulator involved in a wide range of biologic pathways, including liver function and lipid and carbohydrate metabolism.[20] These pathways may relate to SHBG function and sex steroid hormone biology and the development and progression of sex steroid hormone-responsive cancers.[20] At a first glance, *RNLS*, located at another novel locus (10q23.31) shared between AAM and UL, plays no major role in the regulation of AAM or UL. However, this gene encodes a flavin adenine dinucleotide-dependent amine oxidase previously implicated in the growth of cancer cells.[34] Specifically, through the phosphatidylinositol 3-kinases-protein kinase B (PI3K-AKT) and mitogen-activated protein kinase (MAPK) pathways, *RNLS* protects cells by ameliorating ischemia and decreasing cytotoxic effects.[45] A positive correlation between *RNLS* and tumors was demonstrated in pancreatic[23] and breast cancers.[24] Similarly, although the exact function of *R3HDM1* remains unclear, this gene was previously recognized as a candidate oncogene by screening for base-specific mutations.[25] *MiR-128-1*, in an intron in *R3HDM1*, showed a considerable reduced level of expression in human breast cancer specimens.[46] Candidate causal variants and genes identified in our computational analyses will require experimental follow-up for validation.

Integrating data from GWAS and GTEx tissue-specific expression, we investigated genes shared by reproductive traits and UL and if they have potential functional connections with human tissues. Similar to findings from CPASSOC, we found shared genes that are related to cell proliferation (*UIMC1*, *SETD8*, and *ZNF346*),[36,47,48] and cell-cycle and epigenetic regulation (*CDK2AP1*).[28] In addition to tissues of the exocrine and endocrine systems (well-recognized as relevant to reproductive traits and UL),[49,50] our TWAS study and GTEx tissue enrichment analysis consistently reported shared regulatory features in the digestive and cardiovascular systems, suggesting common biology extending beyond hormones. Digestive problems are commonly affected by reproductive factors, including AAM, ANM, and AFB, possibly via hormonal responses[51,52] and/or the gut microbiome.[53] Demonstrated by previous studies, women with premature menopause anticipate a higher risk for endothelial dysfunction, predisposing them to future cardiovascular disease (CVD).[54] As one of the most important risk factors for CVD, hypertension promotes myomatous proliferation (closely related to UL) by a mechanism similar to atheromatous plaque formation.[55] More studies are needed to disclose fully the complex underlying biologic mechanisms.

In addition to substantial pleiotropic effects, our comprehensive 2-sample MR study demonstrated a marked causal relationship between each of the female reproductive traits (AAM, ANM, or AFB) and the risk for UL, largely concordant with findings from previous observational studies that later AAM or later AFB was associated with a lower risk for UL.[56] Although observational evidence of the association between ANM and UL risk remains scarce, our work that leveraged genetic information suggests that later ANM was considerably associated with a higher risk for UL initiation. The estimated causal effects were directionally consistent across a wide range of alternative statistical approaches and sensitivity analyses that are robust for directional pleiotropy and after adjustment for multiple confounders, supporting the validity of findings.

Clinical and research implications

One possible mechanism through which AAM and ANM might influence UL occurrence is via hormone regulation. Early menarche has been associated with early regular onset of ovulation and hence more prolonged exposure to estradiol and progesterone.[57] Women with an early AAM and a late ANM have, on average, an increased number of menstrual cycles and a greater lifelong exposure to ovarian steroid hormones. Estradiol and progesterone induce mature leiomyoma cells to release mitogenic substances that can reach adjacent immature cells and promote their proliferation, thereby supporting UL growth.[58] Furthermore, AAM has also been correlated with higher levels of SHBG.[59] SHBG plays a crucial role in regulating the transport of sex steroids in plasma. As a consequence of its high binding affinity for 17 beta-hydroxysteroid hormones (eg, testosterone and estradiol), elevated levels of circulating SHBG are expected to reduce the overall amount of bioavailable estradiol.[60] In addition, UL may also develop in response to the damage to myometrial cells during menstruation.[61]

Notably, by identifying an independent causal role of AFB in UL after taking into account the effect of AAM, our study suggests other possible pathways besides steroid hormones that link AFB to UL risk. Although the exact protective mechanisms of later AFB on UL risk remain unclear, findings from our MR analysis suggest important public health relevance to women with an early AAM (largely unmodifiable) in that it may be more likely to prevent a higher risk for UL by having a later AFB (largely modifiable). Moreover, by providing genetic evidence for the correlations between UL and AAM, our findings imply that it may be possible to target women with early AAM to prevent UL.

Strengths and limitations

The key strengths of this study include maximizing the combination of accessible data sets, using multiple analytical frameworks to examine the genetic-based associations, and using MR to assess the causal effects of reproductive factors on UL risk. However, several limitations need to be acknowledged. First, caution should be taken with generalizability because the populations included in this study were restricted to women of European ancestry. Further work is needed to investigate whether our findings apply to women from other ancestry groups. Second, despite selecting genetic instruments for which the statistical evidence of association with the exposure is strong, common SNPs only account for a small portion of the total variance in complex traits (which are affected by various components such as genetic, environmental, socioeconomic factors, and their complex interactions). Unfortunately, the design of our study prevents us from considering environmental effects. Third, both the reproductive traits and the UL data sets contained samples from the United Kingdom biobank (sample overlap ranging from 33% to 76% across all traits pairs), which may have an impact on causal inference in MR analysis (although our other genetic approaches were demonstrated to not be subject to bias introduced by sample overlap[11–13]). However, with the sample overlap online program described in the Web resources section,[62] we computed the type 1 error rate attributable to the proportion of overlap between our exposure and outcome GWAS. The results implied that based on the instruments we included, bias imposed by overlapping samples in our statistics is extremely modest (type 1 error rate <0.05). Fourth, we did not investigate UL subtypes based on molecular characteristics or clinical factors (eg, location, multiplicity, and size) because of limited, well-powered GWAS available for these traits. Further subtype-specific GWAS are needed to extend our findings. Fifth, following the completion of our analytical procedure, an updated UL GWAS[63] was published with a sample size of 367,903 (38,660 cases and 329,437 controls). However, univariable, 2-sample MR that used this updated GWAS yielded largely consistent results (Supplemental Figure 5), demonstrating the robustness of our primary findings. Notably, the newly discovered *RNLS* locus in this GWAS coincided with our findings of novel loci in the cross-trait analysis, thereby providing additional support for the other 2 biologically plausible novel genetic associations (ie, *GCKR* and *R3HDM1*). Finally, although we applied data from large-scale GWAS for each trait, additional data from independent cohorts were not available to replicate our findings. Future replication analysis is warranted.

Conclusion

This study furthered our understanding of the previously identified phenotypic associations between female reproductive characteristics (AAM, ANM, AFB) and UL by showing evidence of genetic correlations, shared loci, and causal relationships. Identification of reproductive risk factors at an early stage in the life course of women might facilitate the initiation of strategies to modify potential UL risk. Future research on the mechanistic pathways that underlie the observed genetic relationships is required.

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Additional information

Web resources

Genome-wide association studies (GWAS) summary statistics for age at menarch (AAM) and age at natural menopause (ANM) are publicly available from http://www.reprogen.org/. Age at first birth (AFB) and uterine leiomyomata (UL) GWAS summary statistics (without 23andMe) are publicly available through the National Human Genome Research Institute—European Bioinformatics Institute (NHGRI-EBI) GWAS Catalog https://www.ebi.ac.uk/gwas/downloads/summary-statistics. To request access to 23andMe GWAS summary statistics, please visit https://research.23andme.com/dataset-access/.

The methods used are as follows:

Linkage disequilibrium score regression (LDSC): https://github.com/bulik/ldsc; SUPERGNOVA: https://github.com/qlu-lab/SUPERGNOVA; Cross-Phenotype Association (CPASSOC): http://hal.case.edu/~xxz10/zhuweb/; PLINK: https://www.cog-genomics.org/plink/1.9/; TwoSampleMR: https://mrcieu.github.io/TwoSampleMR/; Variant Effect Predictor (VEP): https://grch37.ensembl.org/info/docs/tools/vep/index.html; 3-Dimensional single-nucleotide polymorphism (3DSNP): http://cbportal.org/3dsnp/; FM-summary: https://github.com/hailianghuang/FM-summary; Coloc: https://chr1swallace.github.io/coloc/; Webgestalt: http://webgestalt.org; functional mapping and annotation of genome-wide association studies (FUMA): https://fuma.ctglab.nl/. Transcriptome-wide associatin studies (TWAS): http://gusevlab.org/projects/fusion/; bias and type 1 error rate for Mendelian randomization with sample overlap:https://sb452.shinyapps.io/overlap/.

Statistical analysis

Global and local genetic correlation analysis

To evaluate a shared genetic basis between reproductive traits and UL, a global genetic correlation analysis was conducted using linkage disequilibrium (LD) score regression (LDSC).[1] LDSC requires only GWAS summary statistics as input, relying on the fact that an effect size estimate for a given SNP aggregates the effects of all SNPs in LD with that of the SNP. The genetic correlation estimate (r_g) ranged from −1 to +1, with −1 indicating a perfect negative correlation and +1 indicating a perfect positive correlation. LDSC analysis was performed using the known LD structure of the European ancestry reference data from the 1000 Genomes Project and was restricted to only HapMap3 SNPs, commonly acknowledged as being well imputed. *P* value threshold of .05 was used to represent statistical significance.

Global genetic correlations estimated by LDSC were based on aggregated information across all variants in the genome. It is possible that although 2 traits are of negligible global genetic correlation, there are specific regions in the genome that contributie to both traits. We therefore calculated pairwise local genetic correlations using SUPERGNOVA.[2] SUPERGNOVA accurately quantifies the similarity between pairs of traits driven by genetic variation in each specific region of the genome using approximately 2353 independent LD blocks with an average length of 1.6 centimorgans. A Bonferroni-corrected *P* value threshold of 0.05/2353 was used to represent statistical significance.

Cross-trait meta-analysis

Shared genetic components suggest that either the genetic variants have an independent effect on both traits (pleiotropy) or the genetic variants influence one trait via its effect on the other (causality). We subsequently performed a cross-trait meta-analysis using Cross-Phenotype Association (CPASSOC)[3] to identify shared loci that affected both traits. CPASSOC provides 2 estimates, namely S_{Hom} and S_{Het}. S_{Hom} is based on a fixed-effect meta-analysis approach that has greater power when genetic effect sizes are homogenous (unlikely to be true when combining multiple traits). As an extension of S_{Hom}, S_{Het} maintains statistical power when genetic effect sizes are heterogeneous, which is more common in practice and was thus adopted for the analysis herein.

Independent shared variants were obtained using PLINK's clumping function[4] by applying the following parameters: –clump-p1 5e-8 –clump-p2 1e-5 –clump-r2 0.2 –clump-kb 500. Within each locus, the variant with the lowest *P* value was kept as index SNP. Significant pleiotropic SNPs were defined as index variants satisfying $P_{CPASSOC} < 5\times10^{-08}$ and $P_{single\text{-}trait} < 1\times10^{-05}$ (for both traits). Novel pleiotropic SNPs were defined as significant pleiotropic SNPs that did not reach genome-wide significance in single-trait GWAS ($5\times10^{-08} < P_{single\text{-}trait} < 1\times10^{-05}$), were independent ($r^2 < 0.20$) of previously reported genome-wide significant SNPs (of reproductive traits and UL), and none of their neighboring SNPs (± 500 kb) reached $P < 5 \times 10^{-08}$ in single-trait GWAS.

We used Ensemble Variant Effect Predictor (VEP)[5] and 3DSNP[6] for detailed functional annotation of the identified pleiotropic SNPs.

Fine mapping credible set analysis

An index SNP does not always represent a causal SNP. We further identified a credible set of variants that were 99% likely, based on posterior probability, to contain causal variants at each of the pleiotropic loci obtained from CPASSOC using FM-summary.[7] Briefly, FM-summary is a Bayesian fine-mapping algorithm that maps only the primary signal and uses a flat prior with steepest descent approximation, assuming that at least 1 causal variant exists within a given region. Details of the method were described elsewhere.[8]

Colocalization analysis

We subsequently performed a colocalization analysis through Coloc[9] to investigate if CPASSOC identified shared loci colocalized at the same causal variant. Coloc uses a Bayesian algorithm to generate posterior probabilities for 5 mutually exclusive hypotheses regarding the sharing of causal variants in a genomic region, namely H_0 (no association), H_1 or H_2 (association to 1 trait only), H_3 (association to both traits, 2 distinct SNPs), and H_4 (association to both traits, 1 shared SNP). We extracted summary statistics for variants within 1000 kb of each shared index SNP and calculated the posterior probabilities under default prior settings. A shared locus was considered colocalized if PPH_4 >0.7.

Pathway and GTEx tissue enrichment analysis

To gain biologic insights into the pleiotropic SNPs identified from CPASSOC, we performed post-GWAS functional annotations using genes annotated by VEP and 3DSNP. We applied the WebGestalt[10] tool to assess enrichment of shared genes in both the Gene Ontology (GO) biological processes and the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. To identify tissues most relevant to shared genes, we performed a GTEx tissue enrichment analysis using the functional mapping and annotation of genome-wide association studies (FUMA)[11] GENE2FUNC process. The Bonferroni correction procedure was used to correct for multiple testing in both analyses.

Transcriptome-wide association study

To identify genes whose expression pattern across tissues implicate etiology or biologic mechanisms shared by reproductive traits and UL, we conducted transcriptome-wide association studies (TWAS) using FUSION[12] by integrating GWAS summary data with expression weights across 48 tissues from the Genotype-Tissue Expression project (GTEx, version 7). We first performed 48 TWASs for each trait and conducted joint or conditional tests for regions with multiple association features.[13] Bonferroni correction was applied to account for multiple comparisons for all gene-tissue pairs tested for each trait (~235,000 gene-tissue pairs in total, significant gene-tissue pairs were defined as $P_{\text{Bonferroni}}$ <.05). We then intersected single-trait TWAS results to examine shared gene-tissue pairs across traits.

Mendelian randomization analysis

Finally, we performed a bidirectional, 2-sample Mendelian randomization (MR) analysis to estimate the causal link between reproductive traits and UL. The inverse-variance weighted (IVW) approach was applied as our primary approach.[14] This method estimates a causal effect across multiple IVs by combining the causal estimates from each IV in a random- or fixed-effect meta-analysis model under the assumption that all IVs are valid instruments or are invalid in such a way that the overall pleiotropy is balanced to be zero. The Cochran's Q statistic was used to evaluate the heterogeneity of individual causal effects. A P value <.05 indicated the presence of heterogeneity, and consequently, a random-effect IVW approach was adopted. Complementary to IVW, we adopted an MR-Egger regression (detects and corrects for bias caused by directional pleiotropy)[15] and a weighted median approach (provides a consistent estimate of the causal effect even when up to 50% of the genetic variants are invalid).[16] Because these 2 approaches have less statistical power than IVW in detecting true causal effects,[17] we defined significant causal associations as results with P values <.05 in the IVW approach and those that showed consistency in the direction of effect in the MR-Egger regression and weighted median approach. To evaluate if genetic liability to UL exerts a causal effect on reproductive traits, we conducted a reverse-direction MR in which UL-associated independent SNPs were used as IVs.

We assessed the validity of the MR relevance assumption[18] by calculating F statistics for each set of our included IVs; all results were >10 (Supplemental Table 5), indicating a negligible risk of weak instrument bias. To further validate the independence and the exclusion restriction assumptions,[18] sensitivity analyses were performed, including (1) an IVW approach with exclusion of pleiotropic SNPs that were associated with potential confounding phenotypes (see details in Supplemental Tables 1−4) according to the GWAS catalog,[19] and (2) a leave-one-out analysis in which 1 IV was removed at a time and an IVW analysis was conducted to re-estimate the causal effect based on the remaining IVs. We additionally repeated the IVW analysis after excluding palindromic IVs (ie, SNPs whose alleles were represented by the same pair of nucleotides on forward and reverse strands that may introduce ambiguity into the identification of the effect allele).

We conducted 2 additional analyses in the framework of multivariable MR (MVMR).[20] To evaluate if the causal relationships observed were confounded by body mass index (BMI), a major risk factor known to influence both reproductive exposures and UL,[21−23] we incorporated BMI with each exposure based on composite IVs involving SNPs for the exposure and 670 (female-specific and genome-wide significant) SNPs for BMI.[24] Moreover, to assess if the effects of reproductive traits were independent of each other, we adjusted each trait for another trait (ie, AAM and ANM, AAM and AFB, ANM and AFB) and combined all 3 traits (AAM, ANM, and AFB) together to evaluate the direct effect of each reproductive exposure. Cochran's Q statistic was computed for each MVMR analysis to assess instrument validity.

All MR analyses were performed using packages TwoSampleMR (version 0.5.6) and MendelianRandomization (version 0.5.1) in software R (version 4.1.3) (R Core Team, Vienna, Austria).

Ethics approval and consent to participate

The GWAS summary statistics used in the current study were aggregated data that did not contain any personal information. The original GWAS obtained ethical approval from the relevant ethics review committees.

Supplemental References

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